

ABELEJE OLANIYI GEORGE

Full-Stack Software Engineer: Real-Time & AI Systems

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Lagos, Nigeria · Open to relocating to San Francisco or remote

SUMMARY

Full-stack software engineer with 5 years of production experience, focused on real-time systems and AI-powered products using Python and Node.js/TypeScript. I architected Truefit.ai, a real-time voice AI platform built on WebRTC and the Gemini Live API. This infrastructure mirrors the low-latency speech capabilities found in adaptive learning and reading-coach products, and I was instrumental in diagnosing and fixing a compound audio pipeline bug that was blocking production readiness. I care about using AI to extend quality instruction to learners who don't have access to it, and I build reliably end to end: backend architecture, async processing, and responsive React interfaces. My work has driven a 91% reduction in API latency and a 58% improvement in system throughput.

TECHNICAL SKILLS

Real-Time & AI: WebRTC (aiortc), Gemini Live API, LLM tool-calling/agent state machines, RAG pipelines, speech/audio pipeline debugging

Languages: Python, Node.js, TypeScript, JavaScript

Backend: FastAPI, Django (DRF), Express-style REST API design, RESTful API design & integration, Webhooks, microservices architecture, async processing

Frontend: React, Next.js, HTML, CSS, Tailwind CSS

Databases: PostgreSQL, MySQL, SQLite, MongoDB (SQL & NoSQL)

Cloud & DevOps: AWS (EC2, S3, RDS, SES), Docker, containerization, CI/CD pipelines, Render, Vercel

Practices: System design & architecture, load & performance testing, API documentation, code review, Git-based workflows, open-source contribution

WORK EXPERIENCE

Full-Stack Developer : Winnov8 (2025 – Present)

- Owned end-to-end development of production features across frontend and backend, delivering scalable, high-performance server-side applications and responsive React interfaces.
- Designed and implemented an intelligent recommendation system, translating product requirements into reliable, testable RESTful services in collaboration with stakeholders.

- Optimized a real-time notification and messaging system, improving throughput by ~58% under live usage through root-cause analysis and performance tuning.
- Participated in architecture discussions and code reviews to ensure code quality, maintainability, and process adherence on a growing production codebase.

Full-Stack Developer : Revela Africa (2025)

- Built core event-management features end-to-end: ticketing, dynamic RSVP, secure payments, and authentication, integrating RESTful APIs with third-party services and databases.
- Integrated Google APIs and payment gateways via webhooks, enabling automated onboarding, calendar syncing, and smooth multi-device checkout.
- Optimized service orchestration with Redis caching to stabilize performance and reduce response times during peak traffic.

Backend Engineer : The ECO Platform (2024)

- Audited and refactored a Django codebase, removing dead code and rewriting inefficient query paths to improve overall API quality and troubleshoot performance issues.
- Reduced API latency by over 91% and improved system throughput to support thousands of concurrent users through systematic performance analysis and root-cause debugging.

Full-Stack Software Engineer : WispTalk Africa (2023 – 2024)

- Led cloud migration to AWS (S3, RDS, EC2, CloudFront), significantly improving uptime and system resilience.
- Strengthened system reliability through improved infrastructure setup, monitoring practices, and incremental performance improvements; contributed to technical documentation.

Freelance Software Developer : Independent (2020 – 2022)

- Delivered data scraping, automation, and web systems end-to-end for clients across industries, independently managing scoping, development, and delivery — entry point into backend engineering and RESTful API development.

PROJECTS

CoopWise : AI-powered group savings & cooperative platform

Stack: FastAPI, PostgreSQL, Redis, Next.js, Smart Contracts (Flow)

- Architected a scalable FastAPI backend handling authentication, contribution cycles, payouts, and real-time financial state for people without collateral or formal credit history.
- Integrated AI agents and RAG pipelines to deliver personalized, user-facing financial insights.
- Implemented blockchain automation and fiat on/off-ramps via Flow; won hackathon recognition for system design, scalability, and real-world impact.

Truefit.ai : Real-time AI voice interview platform

Stack: Python, FastAPI, WebRTC (aiortc), Gemini Live API, PostgreSQL, Redis, Next.js, TypeScript

- Architected a dual-channel real-time system: WebSocket for session signaling, WebRTC/SRTP for all media, to eliminate head-of-line blocking and deliver jitter-resilient audio.
- Diagnosed and fixed a compound audio pipeline bug (a greedy queue drain collapsing 40 audio chunks into one call, combined with a clock reset corrupting frame pacing) that was causing an infinite response loop and blocking shipping.
- Structured the AI agent as a tool-calling, domain-driven state machine with abstract ports, decoupling the agent's logic from the underlying infrastructure so the AI provider (Gemini, OpenAI) can be swapped without rewrites.

Truefit.ai: Real-time AI voice interview platform

EDUCATION

B.Sc. Computer Science : National Open University of Nigeria (Expected December 2026, GPA ~3.58/4.0)

Backend Engineering Professional Program : ALX Africa (2025)